



Pseudo-acoustic anisotropic reverse-time migration of an ocean-bottom cable dataset acquired in the North Sea

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ABSTRACT

To improve the computational efficiency of reverse-time migration (RTM) for vertically transverse isotropic (VTI) media, various acoustic approximations of the elastic wave equations have been presented. Among these, the pseudo-acoustic wave equation, which combines differential and scalar operators, has the advantage that it does not produce shear wave artefacts. In this study, we investigate the feasibility of pseudo-acoustic anisotropic RTM (PA-RTM) for the analysis of synthetic and observed ocean-bottom cable (OBC) datasets of the Volve oil field in the North Sea. To analyse the influence of anisotropic parameters on RTM images and the sensitivity of data components to seismic anisotropy, we perform PA-RTM using synthetic data by incorporating various background models, as well as pressure wavefields and vertical particle acceleration. The synthetic experiments demonstrate that the anisotropic parameter ε plays an important role, whereas δ is negligible in PA-RTM for VTI media, and that vertical particle acceleration is less affected by seismic anisotropy than the pressure wavefields. Our experiments using observed data show that reflectors are better imaged by PA-RTM than by isotropic-RTM, particularly near the reservoir below the strongly anisotropic region, which is supported by angle-domain common image gathers. These results indicate that PA-RTM using only the vertical component is a suitable option for VTI media when insufficient seismic anisotropy data are available.

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Introduction

Reverse-time migration (RTM) based on the two-way wave equation employs multi-pathing to provide accurate, high-resolution images of complex structures, with strong lateral velocity variation (Baysal, Kosloff, and Sherwood 1983; Etgen 1986). In contrast with Kirchhoff migration (Schneider 1978), which was initially used to describe single events due to its high computational efficiency, RTM emerged with the development of superior computational technology and is now widely used in the oil and gas industry.

To locate target reflectors in migrated images correctly, accurate kinematic information of reflected waves is required, which is accomplished with accurate modelling engines. To describe wave propagation in real earth media accurately, the intrinsic characteristics of rocks that affect seismic velocity must be considered. Seismic anisotropy is a physical property that determines seismic velocities for a range of propagation angles (Daley and Hron 1979; Oh and Alkhalifah 2016). Because horizontal velocity is generally faster than the vertical velocity in vertically transverse isotropic (VTI) media, reflectors can be imaged

at the wrong positions when far-offset data are used under the isotropic assumption (Veeken and Moerk-erken 2013).

The behaviour of P- and S-waves in anisotropic media can be accurately described using the elastic wave equations. Although PS data can also provide useful information for locating gas clouds (Li et al. 2001) and can replace weak PP reflections in creating reflector images, the separation of P- and S-waves requires additional cost, which is higher in anisotropic media than in isotropic media (Yan and Sava 2009, 2012). Furthermore, RTM images obtained using the elastic wave equations can be severely contaminated by mode-converted waves, and the computational cost for elastic RTM is higher than that for the acoustic assumption, particularly in 3D imaging.

Acoustic approximations of the elastic wave equations for anisotropic media (i.e. pseudo-acoustic wave equations) have been introduced to compensate for these shortcomings of the elastic wave equations (Alkhalifah 2000). Several studies have been conducted to improve the stability, accuracy, and efficiency of the pseudo-acoustic wave equations (Zhou, Zhang, and

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Bloor 2006; Duveneck et al. 2008; Song and Alkhalifah 2013; Xu and Zhou 2014), which can be classified as mono – or multi-component-based wave equations. The advantages of the mono-component-based wave equation are that it does not produce shear wave artefacts and is computationally efficient in terms of speed and memory allocation, because the single-wave equation is solved to reconstruct the pressure wavefields. Xu et al. (2015) proposed an elliptic decomposition of the quasi-P wave equation to compensate for numerical errors in the direction vectors.

For 2D VTI media, the anisotropic properties can be described using the Thomsen parameters ε and δ (Thomsen 1986). For field data, prior seismic anisotropy information may be unavailable, preventing the correct location of subsurface reflectors, especially for far-offset seismic data. In the present study, we investigate whether the pseudo-acoustic anisotropic (PA-RTM) algorithm for VTI media can be effectively applied to observed ocean-bottom cable (OBC) data. We adopt the mono-component-based pseudo-acoustic wave equation (Xu et al. 2015), which is immune to shear wave artefacts. Prior to our experiment using observed data, we analyse the contribution of anisotropic parameters to RTM and the sensitivity of PA-RTM to various components of synthetic OBC data. Based on this analysis, we perform PA-RTM for several combinations of anisotropic parameters and data components. Finally, we apply PA-RTM to an OBC dataset acquired from the Volve oil field in the North Sea using different background models.

Acoustic anisotropic RTM

Forward problem in 2D VTI media

We adopt the mono-component-based pseudo-acoustic wave equation as a forward modelling operator for the acoustic VTI media (Xu et al. 2015) for its computational efficiency and lack of shear wave artefacts. Through elliptic decomposition of the original pseudo-differential equation (Xu et al. 2015), we obtain the pseudo-acoustic wave equation for VTI media as follows:

$$a_x = \frac{\partial P}{\partial x}, \quad a_z = \frac{\partial P}{\partial z}, \quad (1)$$

$$\frac{\partial^2 P}{\partial t^2} = \left(v_0^2 (1 + 2\varepsilon) \frac{\partial a_x}{\partial x} + v_0^2 \frac{\partial a_z}{\partial z} \right) S_e + f_p, \quad (2)$$

$$S_e = \frac{1}{2} \left(1 + \sqrt{1 - \frac{8(\varepsilon - \delta)n_x^2 n_z^2}{((1 + 2\varepsilon)n_x^2 + n_z^2)^2}} \right), \quad (3)$$

where a_x and a_z are particle acceleration components under the isotropic assumption; P is the pressure wavefields; and v_0 is the vertical P-wave velocity. The Thomsen parameters ε and δ describe the seismic anisotropy; S_e is the elliptic scalar operator; n_x and n_z are unit

vectors of the phase directions; and f_p is the pressure source. Because the physical meaning of the unit vector is the phase direction of the wave front, it can be approximated as the normalised divergence of the pressure wavefields, $\mathbf{n} = (\mathbf{k}/|\mathbf{k}|) \approx (\nabla p/|\nabla p|)$ (Xu and Zhou 2014).

For the numerical simulation, we use the staggered-grid finite-difference method (Graves 1996), where the particle acceleration components are defined between nodal points where the pressure wavefields are assigned. Using a staggered grid to solve the first-order wave equations facilitates the treatment of multi-component data and application of the perfectly matched layer (PML) boundary condition, which is a widely used method for suppressing artificial reflection from the model boundaries in anisotropic modelling. Although the staggered-grid method requires more calculations than the second-order wave equation, the lack of shear wave artefacts makes this method preferable to the single-wave equation.

Imaging conditions

Conventional RTM is based on zero-lag cross-correlation of the source and receiver wavefields (Claerbout 1971) as follows:

$$I(\mathbf{x}) = \int_0^{T_{\max}} W_s(\mathbf{x}, t) W_r(\mathbf{x}, T_{\max} - t) dt, \quad (4)$$

where $W_s(\mathbf{x}, t)$ represents the source (forward) pressure wavefields, and $W_r(\mathbf{x}, T_{\max} - t)$ represents the receiver (backward) pressure wavefields. Both the source and receiver wavefields are obtained by solving Equation (2). The forward wavefield is obtained by propagating waves from a source over time (from 0 to $T_{\max}$), and the receiver wavefield is recorded by propagating observed data at all receivers in reverse time (from $T_{\max}$ to 0). To compensate for dependency on receiver coverage and source wavelets (Chattopadhyay and McMechan 2008), the correlation-based imaging condition can be modified by normalising the source wavefields (Claerbout 1971) as follows:

$$I(\mathbf{x}) = \frac{\int_0^{T_{\max}} W_s(\mathbf{x}, t) W_r(\mathbf{x}, T_{\max} - t) dt}{\int_0^{T_{\max}} W_s(\mathbf{x}, t)^2 dt} \quad (5)$$

RTM images in anisotropic media are affected by anisotropic parameters because they control the group velocity along the wave propagation angle. The influence of anisotropic parameters on P-wave propagation can be described by the ensemble velocity equation (Fomel and Grechka 2001) as follows:

$$v_g^2(\theta) = v_0^2 (1 + 2\delta \sin^2 \theta \cos^2 \theta + 2\varepsilon \sin^4 \theta) \quad (6)$$

where v_g is the group velocity of P-waves and θ is the vertical propagation angle. In Equation (6), the influence of δ on wave propagation is dominant at intermediate reflection angles. Because subsurface anisotropy

is sensitive to far-offset data, the impact of δ on RTM images is weak; however, the influence of ε on the group velocity increases at wide reflection angles and reaches a maximum at 90° . Therefore, the influence of ε is mainly exerted on wave propagation along wide reflection angles and the transmitted wave (i.e. the diving wave). The maximum values of $\sin^2\theta\cos^2\theta$ and $\sin^4\theta$ are 0.25 and 1, respectively, such that the influence of ε on the wave speed is greater than that of δ .

In multicomponent data, the offset range of reflection data used to form RTM images differs according to the type of component (i.e. horizontal or vertical). The horizontal and vertical components measure horizontal and vertical particle motions of the reflected waves arriving at receivers, respectively. Because reflected waves recorded in near-offset data propagate along narrow-angle reflection paths, vertical particle motions are dominant in PP-reflection data recorded at near offsets. In contrast, in far-offset data, reflected waves propagate along wide-angle reflection paths; therefore, horizontal particle motions are dominant in PP-reflection data recorded at far-offsets. Thus, the effects of anisotropic properties can be smaller in vertical

data components than in horizontal components. In isotropic RTM with pressure data, including narrow – and wide-angle reflection data, the stacked image is formed by near – (vertical P-wave velocity), intermediate – (affected by δ), and far-offset data (affected by ε). Therefore, the final stacked image based on an isotropic background model can be contaminated by the inaccurate backpropagation of far-offset data, which are strongly affected by the anisotropic properties of subsurface media. However, for vertical acceleration data mainly containing narrow-angle reflections, the influence of anisotropy is weak, even when an isotropic background model is used and the RTM images are stacked mainly using the vertical P-wave velocity.

Application of PA-RTM to an OBC dataset acquired in the North Sea

Description of the Volve OBC dataset

A 3D OBC survey was conducted in the Volve oil field, located in the Viking graben of the North Sea, in 2010. The acquired seismic data were officially released by

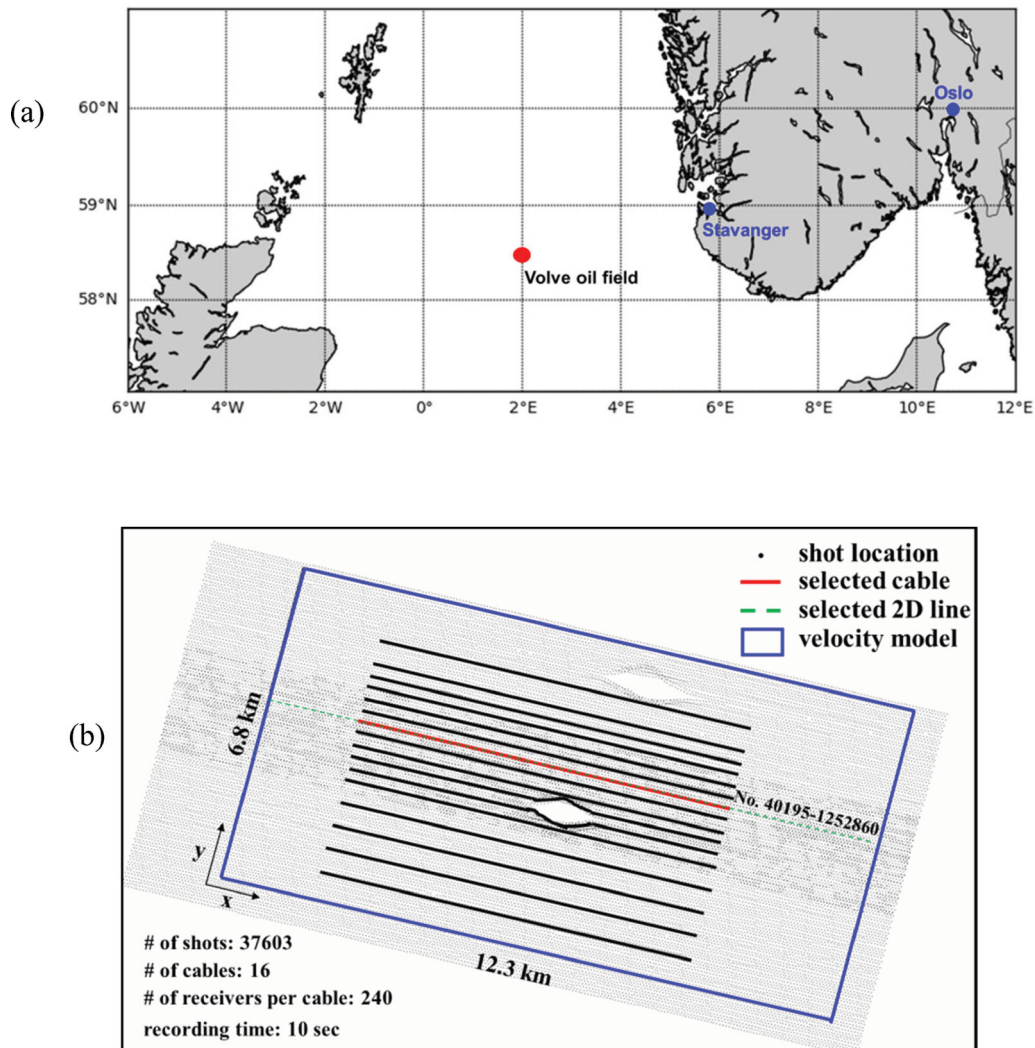


Figure 1. (a) Location of the Volve oil field. (b) Acquisition parameters of 3D ocean bottom cable (OBC) survey in 2010.

Equinor and its former Volve license partners in October 2018 (Figure 1). In total, 37,603 airgun shots were applied, with an interval of 50 m, 16 ocean-bottom cables, and a shot area of 12×2.4 km. Each cable consisted of 240 four-component receivers with an interval of 25 m. The average depth of the receivers was about 80 m and the average recording length was 10 s. The background anisotropic velocity models constructed using reflection tomography were also provided with the dataset created by Equinor and its former Volve license partners. The dimensions of the 3D velocity model were $12.3 \times 6.8 \times 4.5$ km.

The target sandstone reservoir is the Hugin Formation, which has a depth of 3 km and is deposited in the middle Jurassic period, creating a high-velocity chalk layer (Kieft et al. 2010). Well-developed thin layers deposited in the Cenozoic era overlie the caprock; this interbedded thin layering can cause VTI anisotropy

to affect the vertical resolution of the seismic wave, depending on the thickness of the layers (Backus 1962). Therefore, we expect strong VTI anisotropy above the caprock in the Volve oil field, which can significantly degrade the RTM image at reservoir depth.

In this study, we use only one OBC line (red line, Figure 1(b)). The sources are located in the same vertical section as the selected receiver cable. Figures 2 and 3 show the 2D tomography models and their smoothed versions, respectively, which are used as true models to generate synthetic data and PA-RTM initial guesses for VTI media, respectively. Based on these 2D acoustic VTI models, we examine the influence of RTM on anisotropic parameters, particularly below the anisotropic regions, and the sensitivity of data components to seismic anisotropy. We first apply PA-RTM for VTI media to the synthetic data and then conduct a field data experiment.

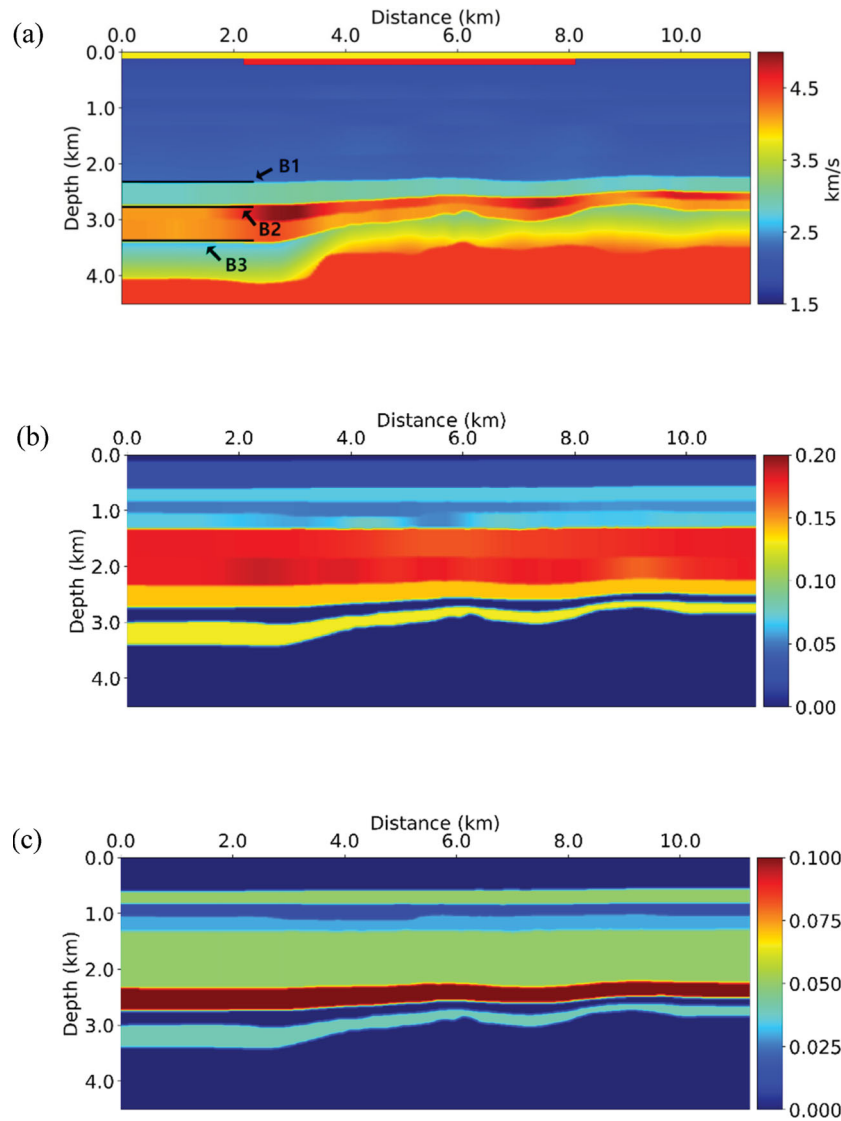


Figure 2. 2D tomography models of the Volve oil field. (a) Vertical P-wave velocity showing the source (the first line) and receiver (the second line), (b) ϵ and (c) δ (provided by Equinor and its former Volve license partners). Black lines indicate the boundaries of layers near the reservoir, which are B1, B2, and B3 from top to bottom, respectively.

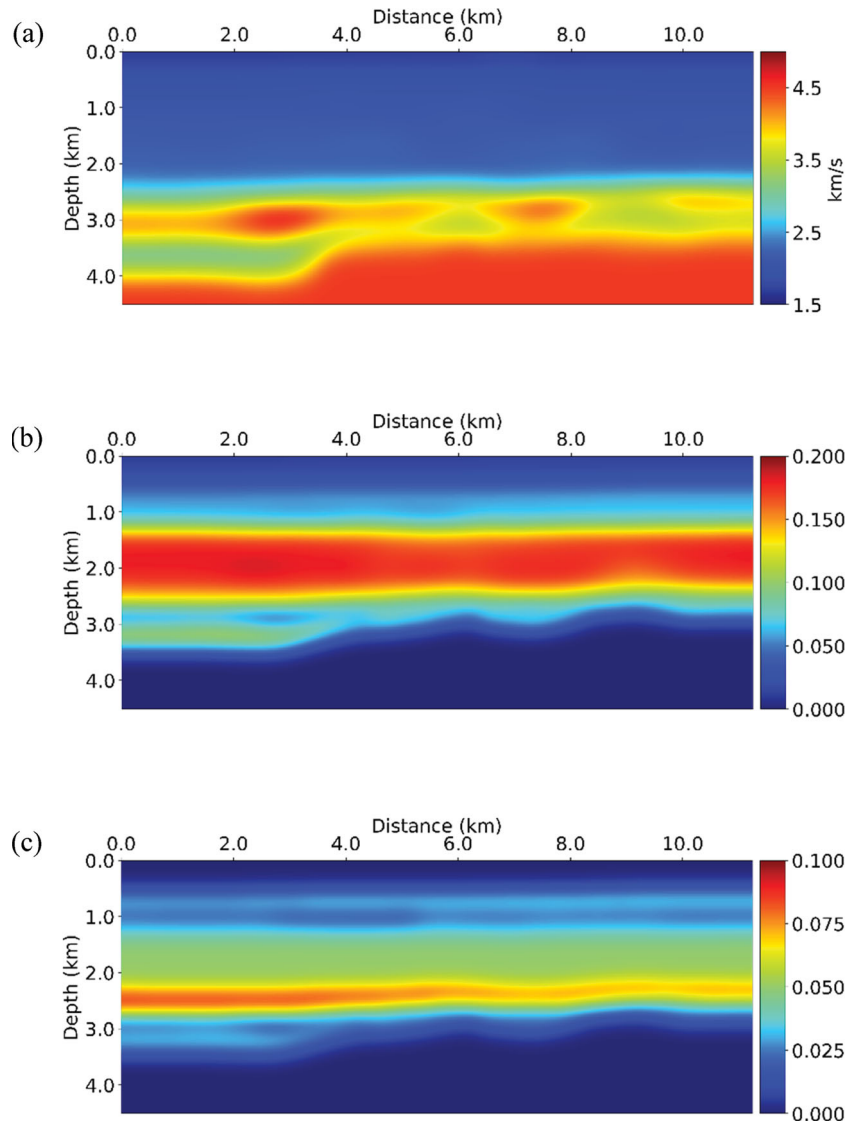


Figure 3. Background anisotropic velocity models for reverse-time migration (RTM) of the Volve data including (a) vertical P-wave velocity, (b) ϵ , and (c) δ .

Anisotropic RTM for synthetic data

To investigate the influence of anisotropic parameters on RTM images with multi-component data, we first apply PA-RTM to the synthetic data generated for 2D VTI media. For the observed data, we use only the pressure wavefields and vertical acceleration; we exclude the horizontal acceleration because horizontal component data are rarely used in RTM for deeper reflectors. We adopt the same acquisition geometry used to acquire the Volve OBC data, with 220 sources and 238 receivers, with intervals of 50 and 25 m, respectively. The migration aperture is 5 km, the recording time was 4 s, and the peak frequency of the Ricker wavelets is 10 Hz. To prevent unwanted migration artefacts, we apply a Laplacian filter to the original image, and then use the SMOOTH2 module in the Seismic Unix software to smooth the original image with a smoothing parameter of 10 on the horizontal axis.

In the tomography velocity model shown in Figure 2(a), we mark the boundaries (B1, B2, and B3) near the

reservoir used for detection in the migrated images. Figure 4 shows RTM results obtained using pressure wavefields for the different background models. The RTM image obtained from the isotropic background model without anisotropy information (Figure 4(a)) is relatively poor, particularly at the boundaries near the reservoir, just below the strongly anisotropic area. Although the isotropic RTM image detects all three boundaries, the reflections overlap and artefacts appear throughout the image, indicating that the layers are not clearly separated. This problem occurs because wave propagation in real media is governed by two different P-wave velocities along the horizontal and vertical directions, whereas wave propagation in isotropic RTM is governed by one representative P-wave velocity. Thus, far-offset data do not correlate well with near-offset data below the anisotropic layers; this is a well-known issue that occurs when isotropic seismic imaging is applied to anisotropic media, commonly referred to as a “hockey stick effect” (Abbasi and Gholami 2018).

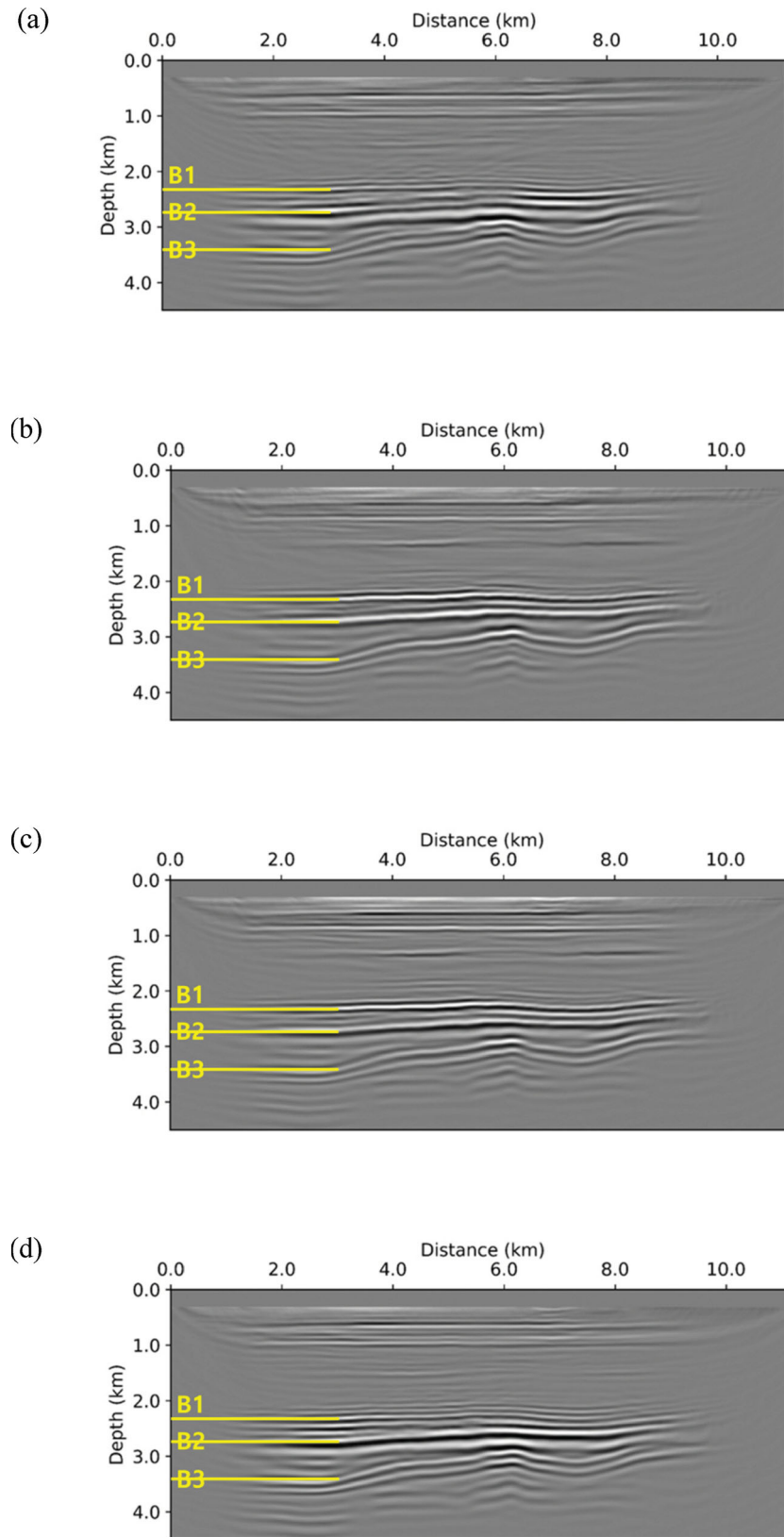


Figure 4. RTM images of synthetic pressure wavefield data. Images were produced using (a) isotropic RTM and anisotropic RTM with (b) ϵ and δ , (c) only ϵ , and (d) only δ . Yellow lines indicate the boundaries of layers near the reservoir.

However, RTM from the background VTI models with all anisotropic parameters clearly recovers the reflector boundaries (Figure 4(b)). The incorporation of anisotropy information allows far-offset data to be migrated to the correct positions, thereby focusing the

correlated image at the reflectors below the anisotropic area. When δ is not included in the background model (Figure 4(c)), RTM still produces a very similar image to that shown in Figure 4(b). When the background media includes only δ (Figure 4(d)), the image is improved

compared to isotropic RTM, but is poorer than other anisotropic RTM results. This finding indicates that the influence of δ on acoustic VTI imaging is not significant

(Alkhalifah, Fomel, and Biondi 2001) and allows us to consider δ negligible in anisotropic FWI. Therefore, we may focus on estimating the seismic velocities and

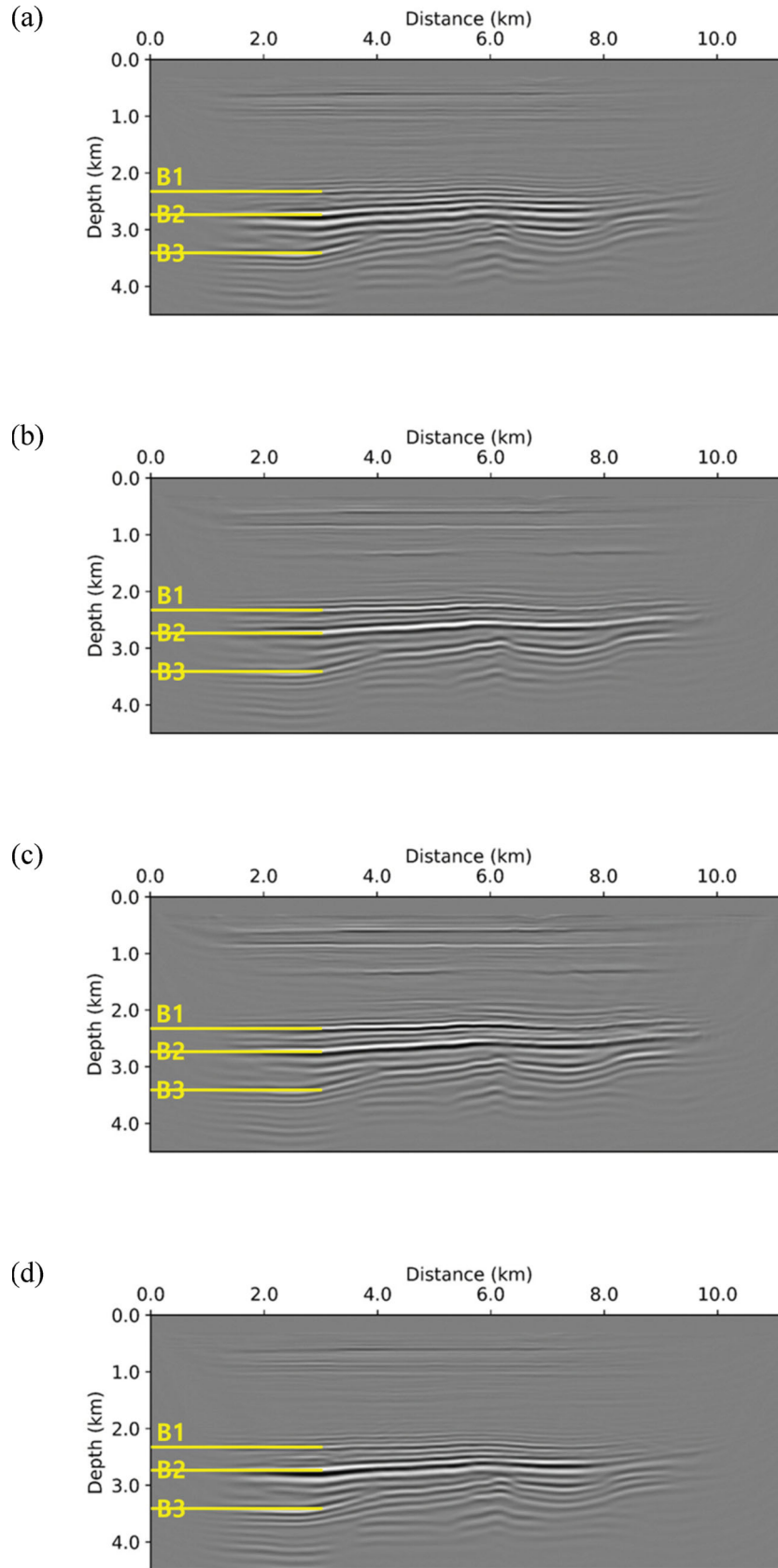


Figure 5. RTM images of synthetic vertical acceleration data. Images were produced using (a) isotropic RTM and anisotropic RTM with (b) ϵ and δ , (c) only ϵ , and (d) only δ . Yellow lines indicate the boundaries of layers near the reservoir.

identifying the anisotropic parameter, ε or η , that determines the ratio of horizontal and vertical velocity.

Figure 5 shows the RTM results obtained using vertical acceleration for different background models. The

influence of subsurface anisotropy is mitigated for the vertical component because the images are generated mainly using near-offset data. The isotropic RTM (Figure 5(a)) shows that the extensions of boundaries

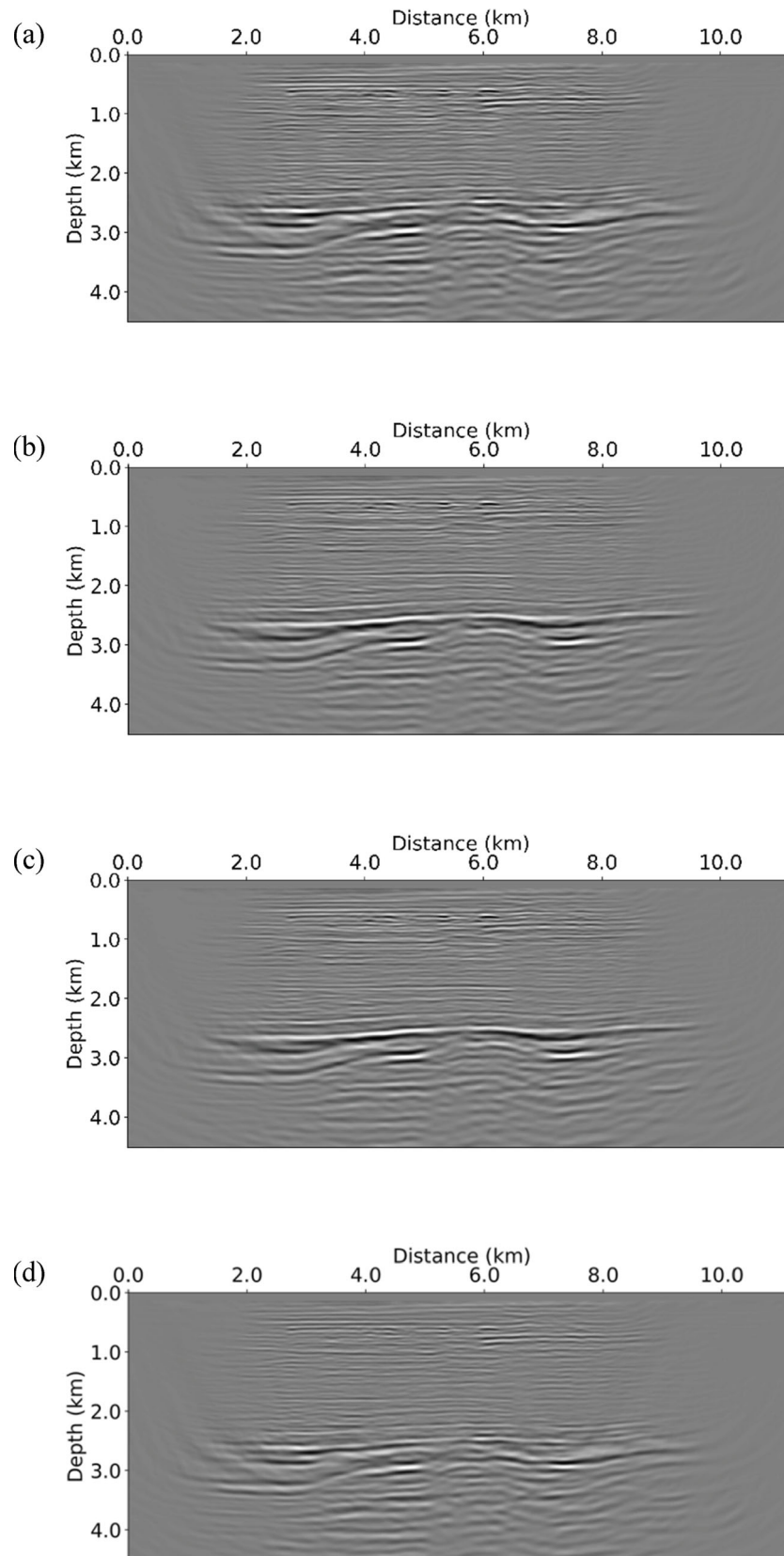


Figure 6. RTM images of observed Volve OBC pressure wavefield data acquired in the North Sea. Images were produced using (a) isotropic RTM and anisotropic RTM with (b) ε and δ , (c) only ε , and (d) only δ .

around the reservoir are inferior to those of the anisotropic RTM; however, the boundaries around the reservoir are distinguishable, with fewer artefacts. Therefore, it is preferable to use the vertical component

of wave motion, which is not sensitive to horizontal wave propagation (mainly affected by VTI anisotropy), when reliable prior seismic anisotropy information is absent.

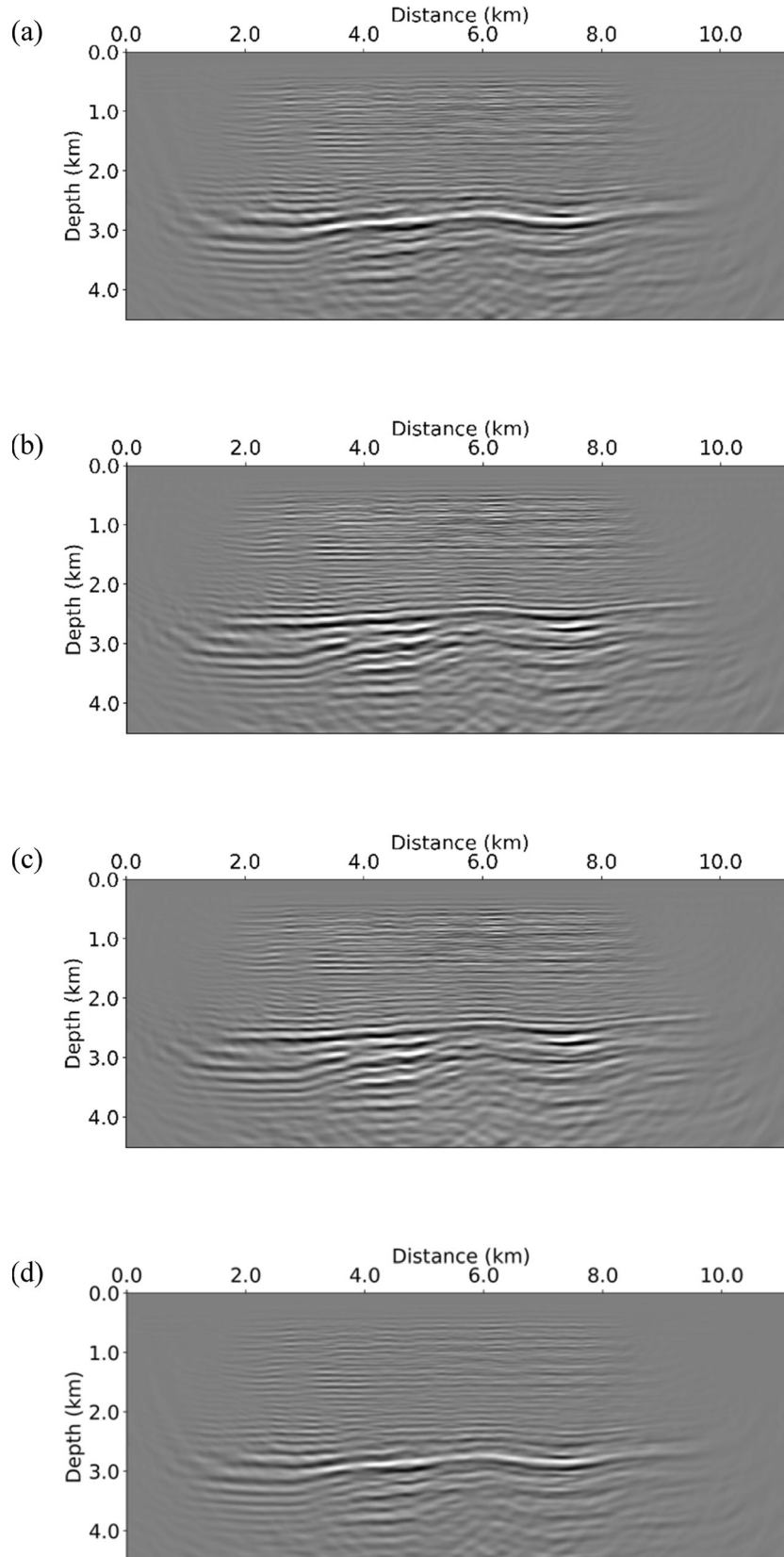


Figure 7. RTM images of observed Volve OBC vertical acceleration data acquired in the North Sea. Images were produced using (a) isotropic RTM and anisotropic RTM with (b) ϵ and δ , (c) only ϵ , and (d) only δ .

Anisotropic RTM for observed OBC data

We pre-process the raw data through direct wave mutation, 3D to 2D conversion (Červený 2005), bandpass filter application, and the exclusion of far-offset data (> 5 km from the source) to reduce migration artefacts. The sources (the first line, Figure 2(a)) are located at the surface and the receivers (the second line, Figure 2(a)) are located at a depth of 80 m. The observed pressure wavefields and vertical acceleration data are used; we exclude horizontal acceleration data as we did for the synthetic test. We assume a 2D anisotropic velocity model with dimensions of 11.25×4.5 km. PA-RTM is conducted with a frequency of 4–25 Hz, 220 shots at an interval of 50 m, and two-component data (pressure and vertical particle acceleration) collected at 238 receivers at an interval of 25 m. We use 4 s of the recorded data. To clarify the migrated image, we use the SMOOTH2 module in the Seismic Unix software and smoothed the original image using a smoothing parameter of 10 on the horizontal axis.

Figure 6 shows RTM images obtained using observed pressure wavefields for different background models. Figure 6(a) shows the image obtained under the isotropic assumption; the locations of the upper

boundaries of the caprock and reservoir are ambiguous and improperly focused due to anisotropic properties. However, despite the inherent instability of the observed data, the migrated reflectors are clearly displayed by acoustic anisotropic RTM, including ε information (Figure 6(b,c)). To directly determine the influence of δ , we implement anisotropic RTM using only δ for the background model (Figure 6(d)). The image is not significantly different from that obtained using isotropic RTM, confirming that δ does not play a role in anisotropic RTM. Compared to the RTM images obtained using synthetic data from the tomography models (Figure 2), thin layers at the shallow part are well recovered using observed data (Figure 6).

Figure 7 shows RTM images obtained using vertical acceleration data for different background models. Due to the intrinsic properties of the vertical component, the isotropic RTM in Figure 7(a) describes the upper boundaries of the caprock and reservoir more clearly than shown in Figure 6(a). The anisotropic RTM images created including ε information are similar (Figure 7(b,c)) and the locations of the target reflectors below the strongly anisotropic area are consistent with those obtained using the pressure wavefield

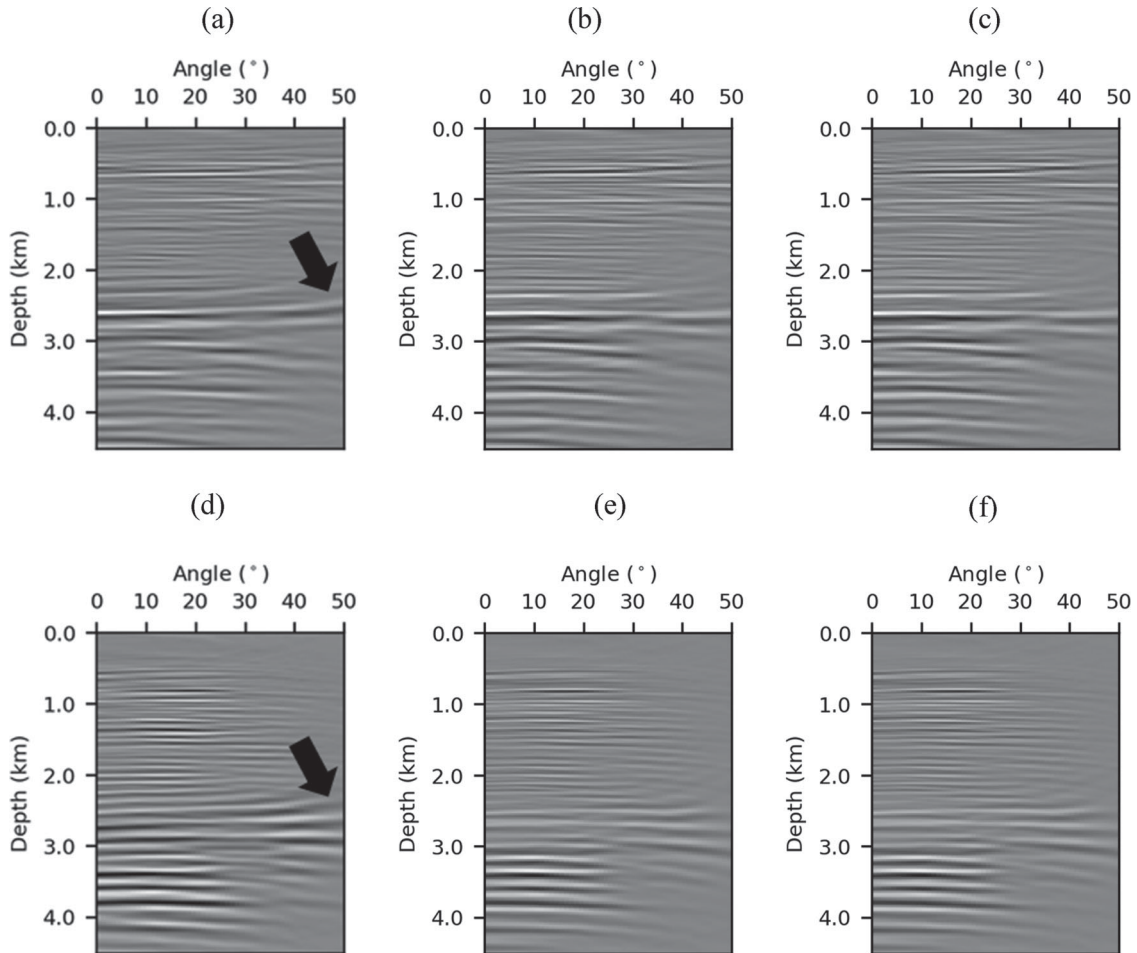


Figure 8. Angle-domain common image gathers (ADCIGs) obtained using pressure wavefield data (a, b, and c) and vertical acceleration data (d, e, and f). Images were produced for a distance of 4 km using (a and d) isotropic RTM and anisotropic RTM with (b and e) ε and δ , (c and f) only ε . Black arrows indicate the “hockey stick effect” at the upper boundary of the reservoir.

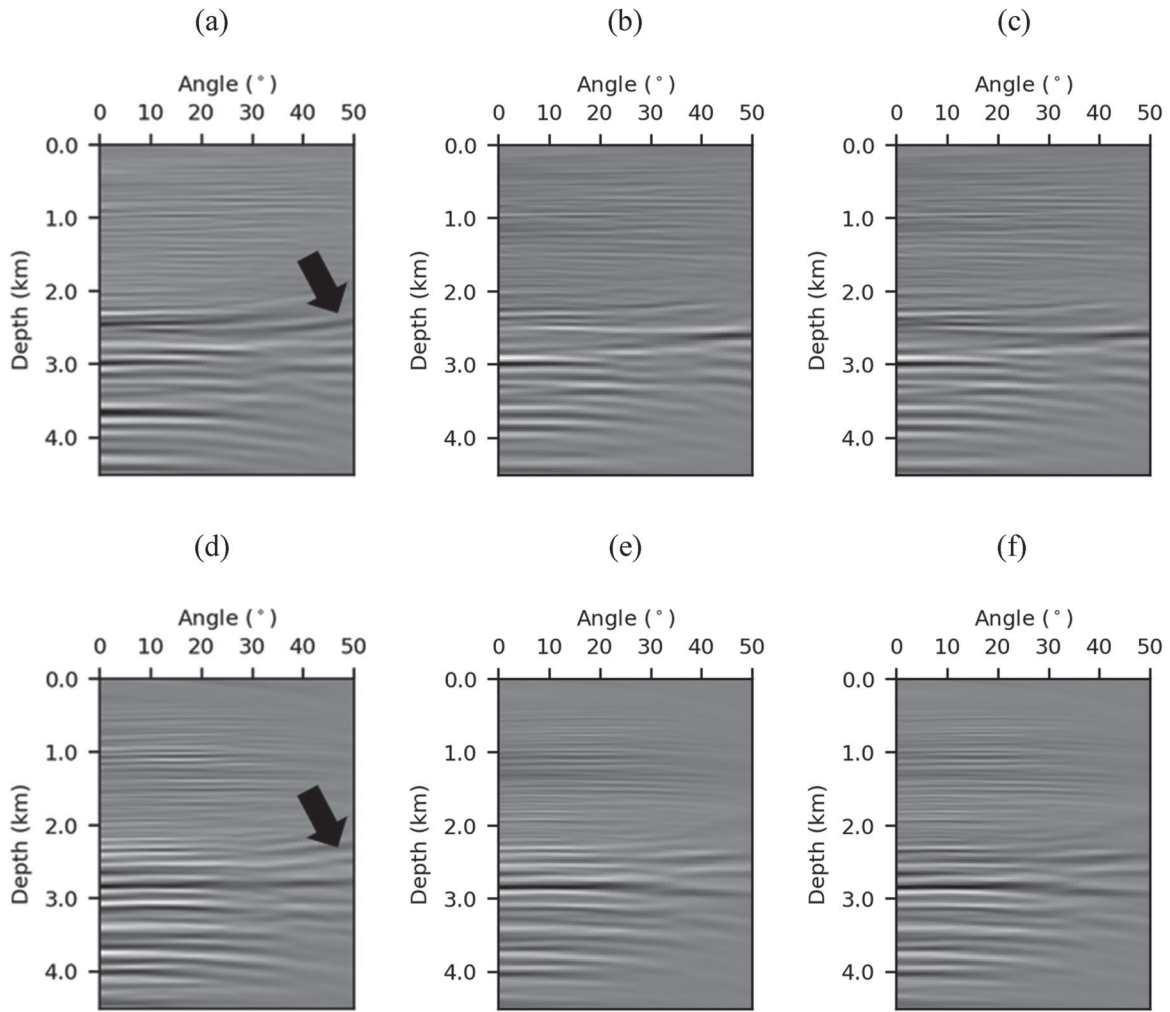


Figure 9. ADCIGs obtained using pressure wavefield data (a, b, and c) and vertical acceleration data (d, e, and f). Images were produced for a distance of 4 km using (a and d) isotropic RTM and anisotropic RTM with (b and e) ϵ and δ , (c and f) only ϵ . Black arrows indicate the “hockey stick effect” at the upper boundary of the reservoir.

data (Figure 6(b,c)). Anisotropic RTM using only δ (Figure 7(d)) produces images that are similar to those produced using isotropic RTM, which is consistent with the results obtained using the pressure wavefield data.

To validate these RTM results, we compute the angle-domain common image gather (ADCIG; Sava and Fomel 2003), which is commonly used for migration-based velocity analysis (Symes 1993). We do not compute the ADCIG of anisotropic RTM using only δ because the migrated images are similar to those produced for isotropic RTM. Figures 8 and 9 show the ADCIG at distances of 4 and 8 km, respectively. Although most reflectors in the images appear flat, we observe hockey stick effects in the ADCIG obtained from the isotropic background model (black arrows, Figures 8 and 9), which we attribute to faster horizontal velocity along the Earth’s surface than vertical velocity. However, Figures 8(d) and 9(d) show that this flatness is not significantly affected by seismic anisotropy when vertical acceleration data rather than pressure wavefield data are used.

Conclusion

In this study, we examined the feasibility of mono-component-based PA-RTM for 2D VTI media using real OBC data acquired in the Volve oil field of the North Sea. To investigate the influence of anisotropic parameters on the RTM image and the sensitivity of data components to seismic anisotropy, we first conducted RTM using synthetic data generated using anisotropic tomography models provided with the observed data. From these synthetic experiments, we confirmed that the anisotropic parameter δ is not a significant factor in PA-RTM for VTI media, which is consistent with results reported in a previous study. Isotropic RTM using only pressure wavefield data located the reflectors poorly compared to anisotropic RTM. However, RTM with vertical acceleration was less affected by seismic anisotropy than that using pressure wavefield data due to the intrinsic properties of vertical acceleration data. Real data examples showed that PA-RTM for VTI media successfully located the target reflectors

(e.g. upper boundaries of the caprock and reservoir below the strongly anisotropic area) using either pressure wavefields or vertical acceleration data, and that vertical acceleration data were less influenced than pressure wavefield data by anisotropic properties. To evaluate the image quality, we generated ADCIGs, and found that most of the reflectors were flatly aligned; these results confirm that it is suitable way to use vertical components and ignore δ in PA-RTM for VTI media, when prior seismic anisotropy information is lacking.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

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